

EXPLOITING UNSTABLE PERIODIC ORBITS OF A CHAOTIC INVARIANT SET FOR SPACECRAFT CONTROL

ELBERT E. N. MACAU

Instituto Nacional de Pesquisas Espaciais – INPE, Caixa Postal 515, 12227-010 São José dos Campos, SP, Brazil, e-mail: elbert@lit.inpe.br

(Received: 3 April 2002; revised: 14 January 2003; accepted: 4 April 2003)

Abstract. The purpose of this work is to show that chaos control techniques (OGY, in special) can be used to efficiently keep a spacecraft around another body performing elaborate orbits. We consider a satellite and a spacecraft moving initially in coplanar and circular orbits, with slightly different radii, around a heavy central planet. The spacecraft, which is the inner body, has a slightly larger angular velocity than the satellite so that, after some time, they eventually go to a situation in which the distance between them becomes sufficiently small, so that they start to interact with one another. This situation is called as an encounter. In previous work we have shown that this scenario is a typical situation of a chaotic scattering for some well-defined range of parameters. Considering this scenario, we first show how it is possible to find the unstable periodic orbits that are located in the chaotic invariant set. From the set of unstable periodic orbits, we select the ones that can be combined to provide the desired elaborate orbit. Then, chaos control technique based on the OGY method is used to keep the spacecraft in the desired orbit. Finally, we analyze the results and make considerations regarding a realistic scenario of space exploration.

Key words: chaos, periodic orbits, control, encounter, OGY

1. Introduction

Poincaré (1889) at the end of the 19th century studying the classical three-body problem concluded that sometimes the motion can be very complicated and apparently unpredictable. This kind of behavior is now called chaotic, and can occur commonly in a large variety of natural systems, even in very simple nonlinear deterministic systems, like a forced pendulum. In fact, chaos can be present in nonlinear, continuous-in-time, dynamical systems with more than one degree of freedom (Arrowsmith and Place, 1994). More elaborate systems where chaos was observed are the Rayleigh-Bénard convection (Libchaber and Maurer, 1982), the Arechi optical experiment (Arechi et al., 1982), the Belousov-Zhabotinsky chemical reaction (Hudson and Mankin, 1981), among others.

The main characteristic of a chaotic behavior is that its time evolution presents sensitive dependence on initial condition on a closed invariant set (Arrowsmith and Place, 1994). In long time, the consequence of this property, which is known as the *butterfly effect*, is the unpredictability of the state of the system. Sensitive



dependence on initial condition can also be exploited to alter the system's behavior by using small perturbation (Ott et al., 1990; Shinbrot et al., 1990). Previous works have demonstrated this principle both theoretically (Bollt and Meiss, 1995, 1996; Macau, 1999, 2000) and in real situations (Belbruno and Miller, 1993; Belbruno, 1994) in guiding spacecrafts. On this regard, it is particularly relevant to mention low energy transfers to the Moon using *weak capture*. In 1991, the spacecraft Hiten reached the Moon following a transfer constructed from this approach. This methodology, exploit a region in phase space that support a special type of chaotic motion for initial conditions on it and was termed *weak stability boundaries* (Belbruno, 2002a). These transfer trajectories (named ballistic capture) are low energy since capture at the Moon into an elliptical orbit is automatic and does not require braking rockets (Bello-Mora et al., 2000), which contrast with the traditional Hohmann transfer. A proof of the existence of a chaotic invariant set associated with this unstable ballistic capture has recently been obtained (Belbruno, 2002b, 2003).

Control of chaos is another milestone that advantageously uses the butterfly effect. Ott et al. (1990) exploited the sensitive dependency on initial condition of the chaotic evolution to stabilize chaotic trajectory about a periodic orbit. Their approach, which is known as OGY strategy of chaos control, uses only small perturbations applied to control parameters to stabilize chaos, keeping the parameters in a neighborhood of their nominal values. The feasibility of the OGY method was experimentally demonstrated in many experiments (Ditto et al., 1990; Azevedo and Rezende, 1991; Garfinkel et al., 1992).

Besides using the butterfly effect, the OGY method exploit another remarkable property of the chaotic invariant set: It has embedding on it an infinite, numerable and dense set of unstable periodic orbits (Arrowsmith and Place, 1994). Thus, to apply the OGY method, first it is necessary to identify one of these unstable periodic orbits. As there are practically many of these orbits, the one to be chosen can be specifically the one to yield a desirable system performance. After choosing the desired unstable periodic orbits, small perturbations can be used to drive the chaotic trajectory to it as well as to keep the trajectory about the selected unstable periodic orbit. The breakthrough of the OGY's approach is to show that one can better control chaos by taking advantage of chaos (Grebogi and Lai, 1997). In applications in which chaos is unavoidable, there are many unstable periodic states that can be stabilized to yield a desirable system performance.

Although the sensitivity dependence on initial condition of a chaotic system has already been used in astrodynamics both in real (Belbruno and Miller, 1993; Belbruno, 1994) and simulated missions (Macau, 1999, 2000; Bollt and Meiss, 1995, 1996), the same is not true in regard to the use of the dense set of unstable periodic orbits embedded on the chaotic invariant set. In astrodynamics, its is common to face situations where the goal is to stabilize the orbit of a spacecraft about a satellite. In this paper we show that if chaos is present, situations like that can be efficiently accomplished by using the OGY's approach. Furthermore, as the set of

unstable periodic orbits are dense, infinite and numerable in the chaotic invariant set, it is possible to have periodic orbits with almost all desirable shapes and characteristics so that these orbits can support a large variety of mission requirements for space exploration. Besides, targeting techniques (Shinbrot et al., 1990; Macau, 1998, 1999) can be used to combine these orbits resulting an almost unlimited source of elaborate orbits.

The organization of the paper is as follows. In the next section we show how the unstable periodic orbits can be found in a Hamiltonian chaotic system. In Section 3 we review the basic theory of the OGY method and present a formulation for parameter perturbations that does not explicitly involve eigenvalues. In Section 4 we consider a situation of interest for the astrodynamics characterized by a satellite and a spacecraft moving initially in coplanar and circular orbits, with slightly different radii, around a heavy central planet, which is described by Hill's equation. In Section 5 we applied the OGY method to this problem. In Section 6, we present the conclusion.

2. Detecting Unstable Periodic Orbits

Unstable periodic orbits (UPOs) are the most fundamental building blocks of a chaotic system (Arrowsmith and Place, 1994). A chaotic invariant set has embedded in it an infinite number of UPOs. These orbits provides the skeleton of the chaotic invariant sets, and many physical quantities of these systems, as the case of Lyapunov exponents, natural measure, fractal dimension, can be computed from them (Cvitanovic and Eckhardt, 1989; Auerbach, 1990; Cvitanovic, 1992). Furthermore, knowing the position of the UPOs is also fundamental for the application point of view, as there location must be know to establish an partition suitable for application in communication as well as is the first step in applying control of chaos technics (Davidchack and Lai, 1999). However, the determination of the UPOs embedding in the chaotic invariant set is not a trivial task (Schmelcher and Diakonov, 1998). Although the number of UPOs grows exponentially as the period increases at a rate given by the topological entropy, and they are dense in the chaotic invariant set, they are of measure zero, whereas the chaotic orbits are of measure one. To face this difficult task, a number of recent works have been proposed in the seeking for an efficient method for the detection of UPOs (Biham and Wenzel, 1989; Grassberger et al., 1989; Hansen, 1995; Davidchack and Lai, 1999).

Among those methods, Schmelcher and Diakonov proposed one (Schmelcher and Diakonov, 1998) that is very simple to be implemented and, for the first time, allows the systematic computation of UPOs of high period, in special for chaotic maps. It is based on a globally convergent iterative scheme that, according to the authors, converges to UPOs from any initial point. However, as pointed out in Davidchack and Lai (1999), it is impossible to assure that all the UPOs are found

using it, and it is not very efficient in related to convergence. Despite that, we decided to use it in this work because of its remarkable simplicity concerning computational implementation and efficiency. Furthermore, in general, it finds the UPOs are very fast.

Let us explain the method using a N -dimensional chaotic map given by

$$U : \mathbf{X}_{i+1} = \mathbf{F}(\mathbf{X}_i). \quad (1)$$

The goal is to find for U its unstable fixed points (FPs). Note that periodic orbits of period p are fixed points of $\mathbf{F}^{[p]}$. Thus, the same method for finding fixed point can also be used, in principle, to find periodic orbits of any period. To find the FPs for U , the method constructs from U a transformed dynamical system (S) which poses FPs at the same positions as U but instead of being unstable, have become stable in the transformed system. If this operation can be accomplished, each trajectory of S after some iterations closely approaches a FP of U . Therefore, the FPs of U can be found by using as initial conditions a uniform grid in the domain of S , for example.

The following transformed map S satisfies the requirements in related to the one to one correspondence between its FPs and the ones of U :

$$S_k : \mathbf{X}_{i+1} = \mathbf{X}_i + \mathbf{W}_k[\mathbf{F}(\mathbf{X}_i) - \mathbf{X}_i], \quad (2)$$

where $\mathbf{W}_k$ is a constant invertible $N \times N$ matrix. The next step is to fulfill the requirement of having the FPs of S stable. The authors used the following argument to show that it is always possible to find a matrix $\mathbf{W}_k$ for this purpose. The Jacobian matrix for S and U are related by

$$\mathbf{M}_{S_k} = \mathbf{1} + \mathbf{W}_k(\mathbf{M}_U - \mathbf{1}). \quad (3)$$

For an unstable fixed point $\mathbf{X}_f$, at least one of the eigenvalues of $\mathbf{M}_U$ at $\mathbf{X}_f$ must have an absolute value greater than 1. $\mathbf{W}_k$ can be written for each of its elements as $(\mathbf{W}_k)_{ij} = (\lambda \mathbf{C}_k)_{ij}$, with $1 \gg \lambda > 0$ and $(\mathbf{C}_k)_{ij} = O(1)$. To stabilize $\mathbf{X}_f$ in S_k , first $\mathbf{C}_k$ has to be chosen such that the real parts of all eigenvalues of $\mathbf{C}_k(\mathbf{M}_U - \mathbf{1})$ are negative. If this is achieved, the authors argued (Schmelcher and Diakonov, 1998) that it is always possible to find a small value for the parameter λ such that the Jacobian matrix $\mathbf{M}_{S_k} = \mathbf{1} + \mathbf{W}_k(\mathbf{M}_U - \mathbf{1})$ evaluated at $\mathbf{X}_f$ has the real parts of all eigenvalues less than 1, which implies that $\mathbf{X}_f$ is a stable fixed point for S_k . Subsequently, it is shown that there is a restrict set of matrices $\mathbf{C}_k$ that fulfill the requirements. All the elements of this set are involutory matrix, that is, $\mathbf{C}_k^2 = \mathbf{1}$, and corresponding to special reflections and rotations in space. The elements of this set of matrices $\{\mathbf{C}_k\}$ are $(\mathbf{C}_k)_{ij} \in \{0, \pm 1\}$, and each row or column contains only one element which is different from zero. For a N -dimensional space, the number of such matrices is given by $N! \times 2^N$. Thus, to obtain all the FPs, the state space of S has to be explored for each one of these matrices.

We have said that this method is not very efficient for the standpoint of convergence. However, the solution given from it can be improved by using the traditional

Newton–Raphson (NR) method to find the fixed points for U , using as initial conditions the solutions obtained by the Schmelcher–Diakonov (SD) method. Near the fixed point, the NR has excellent convergence properties, provided that the initial point is within the linear neighborhood of the solution.

These combined method can also be used for continuous-in-time dynamical systems with the introduction of a Poincaré section transversely to the trajectory flow. If this Poincaré section is properly defined, we claim the SD method combined with the NR method it is also a very efficient tool to find unstable periodic orbits for continuous-in-time chaotic dynamical systems (Roth, 2003).

3. Stabilizing Chaotic Trajectories

One fundamental aspect of chaotic invariant sets is that they are typically permeated by an infinite dense set of unstable periodic orbits. Control of chaos by stabilizing those unstable periodic orbits was conceived by Ott et al. (1990) (OGY). Their basic ideas regarding a chaotic attractor is as follows. First one chooses an unstable periodic orbit embedded in the chaotic set according to some performance criterion. Second, one defines a small region around the desired periodic orbit. As a chaotic trajectory is transitive in its invariant set, starting from any initial condition, after some time it will come to that small region about the chosen unstable periodic orbit. When this occurs, small judiciously chosen parameter perturbations are applied to force and keep the trajectory evolving on the unstable periodic orbit. In subsequent works, Lai et al. (1993a, b) showed that with small modifications the OGY method can be applied to hyperbolic Hamiltonian systems, as follows.

Let us consider the above discrete time dynamical system,

$$\mathbf{X}_{i+1} = \mathbf{F}(\mathbf{X}_i, p), \quad (4)$$

where $\mathbf{X}_i \in \mathfrak{N}^2$, $p \in \mathfrak{N}$ is an externally controllable parameter, and $\mathbf{F}$ is a smooth function in both variables. The parameter perturbation to be used to control the system is required to be small, that is, it is required that

$$|p - p_0| < \delta, \quad (5)$$

where p_0 is some nominal parameter value and δ is a small number defining the range of parameter variation. The goal is to program the parameter p in such a way that a typical trajectory in the chaotic region is stabilized about the desirable unstable periodic orbit. The stabilization procedure just start to actuate when the chaotic trajectory enters a small region around one of the periodic-orbit point, whose size is proportional to δ . Once the particle is inside this small region, p is judiciously changed to keep the trajectory around the unstable periodic orbit.

Let we assume the UPO of period m to be controlled is

$$\mathbf{X}_{01}(p) \rightarrow \mathbf{X}_{02}(p) \rightarrow \cdots \rightarrow \mathbf{X}_{0m}(p) \rightarrow \mathbf{X}_{0(m+1)}(p) = \mathbf{X}_{01}(p). \quad (6)$$

The linearized dynamics in the neighborhood of the period- m orbit is

$$\mathbf{X}_{n+1} - \mathbf{X}_{0(n+1)}(p_n) = \mathbf{M}[\mathbf{X}_n - \mathbf{X}_{0n}(p_n)], \quad (7)$$

where $\mathbf{M}$ is the two-dimensional Jacobian matrix at the orbit point $\mathbf{X}_{0n}$, $p_n = p_0 + (\Delta p)_n \mathbf{g}_n$ where from Equation (5) $(\Delta p)_n < \delta$. Observe that the parameter variation will result in the following change on the periodic-orbit points:

$$\mathbf{X}_{0n}(p_n) - \mathbf{X}_{0n}(p_0) \approx (\Delta p)_n \mathbf{g}_n, \quad (8)$$

where $\mathbf{g}_n = \partial \mathbf{X}_{0n}(p) / \partial p |_{p_0}$.

The Jacobian matrix $\mathbf{M}$ of Equation (7) can be expressed in terms of its stable and unstable directions. Note that even in the case in which $\mathbf{M}$ has complex-conjugate eigenvalues, the stable and unstable directions for $\mathbf{M}$ can be defined. To geometrically find the stable direction at a point $\mathbf{X}_0$, we first iterate this point forward N times under the map $\mathbf{F}$ and get the trajectory $\mathbf{X}_1 = \mathbf{F}(\mathbf{X}_0)$, $\mathbf{X}_2 = \mathbf{F}(\mathbf{X}_1) = \mathbf{F}^{[2]}(\mathbf{X}_0)$, $\dots$, $\mathbf{X}_N = \mathbf{F}(\mathbf{X}_{N-1}) = \mathbf{F}^{[N]}(\mathbf{X}_0)$. Now, a circle of arbitrarily small radius ε is put at the point $\mathbf{X}_N$. If this circle is iterated backward once, the circle will become an ellipse at the point $\mathbf{X}_{N-1}$ with the major axis along the stable direction of the point $\mathbf{X}_{N-1}$. This ellipse is iterated backward, while at the same time its major axis is kept of order ε via certain normalization method. This procedure is repeated all the way back to the point $\mathbf{X}_0$, where the ellipse becomes very thin, with its major axis along the stable direction provided N is large enough. This procedure is schematically shown in Figure 1.

Similarly, as shown in Figure 2, to find the unstable direction at point $\mathbf{X}_0$, first this point is iterated backward under the inverse map N times to get a backward orbit $\mathbf{X}_{-j} = \mathbf{F}^{[-j]}(\mathbf{X}_0)$ ($j = 1, \dots, N$). In the point $\mathbf{X}_{-N}$ a circle of arbitrarily small radius ε is put at it. After that, the previously discussed procedure is applied,

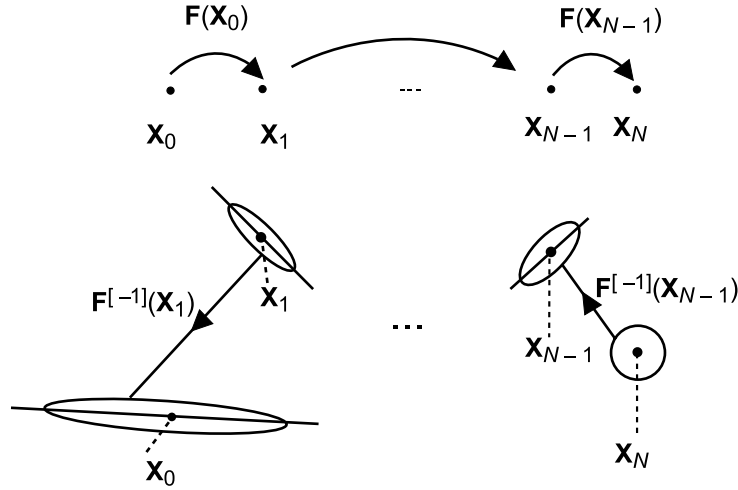


Figure 1. Geometric approach to find the stable direction associated with the point $\mathbf{X}_0$.

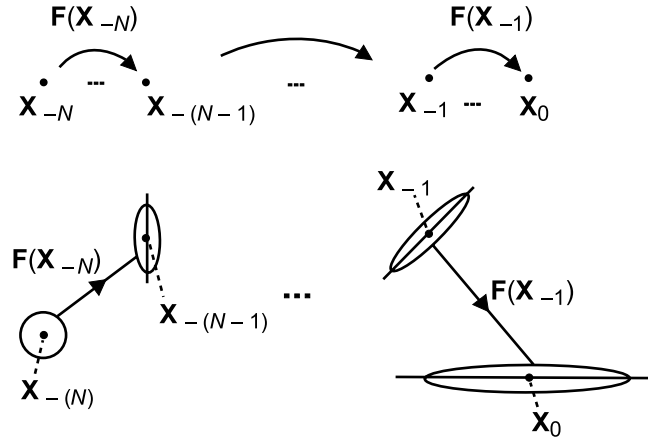


Figure 2. Geometric approach to find the unstable direction associated with the point $\mathbf{X}_0$.

but now iterating in the forward direction. At the end of this procedure, in $\mathbf{X}_0$ the ellipse becomes very thin, with its major axis along the unstable direction.

Let $\mathbf{e}_{s(n)}$ and $\mathbf{e}_{u(n)}$ be the stable and unstable directions at $\mathbf{X}_{0n}$, and let $\mathbf{f}_{s(n)}$ and $\mathbf{f}_{u(n)}$ be two vectors that satisfy $\mathbf{f}_{u(n)} \cdot \mathbf{e}_{u(n)} = \mathbf{f}_{s(n)} \cdot \mathbf{e}_{s(n)} = 1$ and $\mathbf{f}_{u(n)} \cdot \mathbf{e}_{s(n)} = \mathbf{f}_{s(n)} \cdot \mathbf{e}_{u(n)} = 0$. To control the orbit, it is required that the next iteration of a trajectory point after falling into one of the small neighborhoods around $\mathbf{X}_{0n}$ lies on the stable direction at $\mathbf{X}_{0(n+1)}(p_0)$, that is,

$$[\mathbf{X}_{n+1} - \mathbf{X}_{0(n+1)}(p_0)] \cdot \mathbf{f}_{u(n+1)} = 0. \quad (9)$$

Substituting Equations (7) and (8) into Equation (9), we obtain the following expression for the parameter perturbations:

$$(\Delta p)_n = \frac{\{\mathbf{M}[\mathbf{X}_{n+1} - \mathbf{X}_{0(n+1)}(p_0)]\} \cdot \mathbf{f}_{u(n+1)}}{[(\mathbf{M}\mathbf{g}_n) - \mathbf{g}_{n+1}] \cdot \mathbf{f}_{u(n+1)}}, \quad (10)$$

where $\mathbf{M}$ is evaluated at $\mathbf{X}_{0n}(p_0)$. This parameter perturbation is applied at each time step of the trajectory, which is so kept stabilized around the UPO.

4. Hill's Equations

We now consider a situation of interest for the astrodynamics. We consider two light bodies M_2 and M_3 describing initially coplanar and circular orbits, with slightly different radii, around a heavy central body M_1 . If the two light bodies are far apart, their mutual attraction can be neglected, and the problem reduces in a fair approximation to a superposition of two independent two-body problems. However, if the distance between them becomes sufficiently small, a situation called as an *encounter* happens, in which their mutual attraction is no longer negligible and cannot be ignored. This is known as *Hill's problem* (Hill, 1878).

The Hill's problem can be considered as the simplest non-integrable case of the N -body problem. The equations are very simple, and contain no parameter. However, many problems in celestial mechanics can be adequately approximated by Hill's equations. Examples are: the Sun–Earth–Moon problem (Hill, 1878), which was the motivation of Hill's original work; the interaction between particles in planetary rings (Goldreich and Tremaine, 1979, 1980; Hénon, 1981); the motion of coorbital satellites (Dermott and Murray, 1981); the accretion of particles by a proto-planet (Schofield, 1981); the distribution of particles around the Earth (Dole, 1962). Hill's problem is applicable to situations where one mass (primary) is much larger than the two others (secondary and the third smaller body). In relation to the restrict three-body problem, the Hill's problem can be considered as a simplified or a limit version of that one where the distance between the two massive primaries is made very large so that a neighborhood of the secondary can be studied in details. Actually, the Hill's equation can be derived from the circular restrict three-body problem by scaling the mass of the secondary to 1 as the primary is sent to an infinity distance and has its mass also increased to infinity, so that its action can be regard as a force field on a given direction (Simó and Stuchi, 2000).

In the present paper we consider the Hill's problem for the case where M_2 is a natural satellite of M_1 , while M_3 is an artificial satellite and our goal is to stabilize the later in a convenient orbit about M_2 . The difference between the radii of their initial circular orbits is called the *impact parameter*, and represented by the parameter h . In the initial situation, M_2 , the inner body, has a slightly larger angular velocity than M_3 , which means that after some time they eventually go to an encounter. In Macau (1999), we have analyzed the dynamics of this problem, showing that there are region of the impact parameter where a chaotic scattering appears. In this regions, it is possible to stabilize the chaotic evolution of the satellites around one of the UPOs of the chaotic evolution using the previously discussed method.

Let us assume that the mass of either satellite is small compared to the mass of the planet, that is,

$$m_1 \gg m_2 \quad \text{and} \quad m_1 \gg m_3, \quad (11)$$

where m_i is the mass of body M_i . The ratio of the two masses m_2 and m_3 can be arbitrary, but fixed. Thus, we define

$$m = m_1 + m_2 + m_3, \quad (12)$$

and

$$\mu = \frac{m_2 + m_3}{m_1}. \quad (13)$$

We consider that the distance between the two satellites is small compared to their distance to the planet. Thus, the two satellites can be approximately view as a single body in orbit around the planet. We call this orbit as the *mean orbit* (Hénon and

Petit, 1986; Petit and Hénon, 1986), and we assume it is a circular orbit with radius a_0 . The angular velocity of m_2 and m_3 on this mean orbit is

$$\omega = \sqrt{Gma_0^{-3}}. \quad (14)$$

In Hénon and Petit (1986) and Petit and Hénon (1986) it is shown that when the satellites are moving in the neighborhood of each other and the movement is described using synodic coordinates, a sequence of transformation can be applied, resulting in the following equations for the relative motion of the satellites described in ξ and η coordinates, which is called *Hill's equations*:

$$\dot{\xi} = u, \quad \dot{\eta} = v, \quad \dot{u} = 2v + 3\xi - \frac{\xi}{\rho^3}, \quad \dot{v} = -2u - \frac{\eta}{\rho^3}, \quad (15)$$

where

$$\rho = \sqrt{\xi^2 + \eta^2}. \quad (16)$$

Hill's equations admit the integral

$$\Gamma = 3\xi^2 + \frac{2}{\rho} - u^2 - v^2 \quad (17)$$

which is called the *Jacobi integral* by analogy with the restrict problem. The Jacobi integral can be written in terms of the initial conditions (impact parameter)

$$\Gamma = \frac{3}{4}h^2. \quad (18)$$

The problem that we are considering is typically characterized by having three well-defined phases. In the first phase, the satellites approach from each other; in the second phase, the two satellites remain close to each other and a complex relative motion takes place; in the last phase, the two satellites move away from each other. Thus, this problem is a typical case of a classical *scattering*.

For large values of the parameter h , the orbit of the lightest satellite is only slightly perturbed, as can be seen in Figure 3. As the parameter diminishes, the perturbation increases and a loop appears. After that, the shape of the orbit changes rapidly with the parameter. Continue to decrease the value of the parameter, we find an interval in which the shape of the orbits changes wildly. Inside this interval, there is an extremely high sensitivity of the shape of the orbit to just small changes in the parameter. After this interval, as the parameter is reduced, the shape of the orbit changes continuously and slowly with the parameter. Then a new interval of rapidly changes of the shape of the orbit with the variation of the parameter appears, after which follows another interval of wildly variation. This succession of behaviors continues to happen as the parameter is diminished, until the appearance of horseshoe orbits.

Figure 4 represents the *scattering function*, which is the final impact parameter h' as a function of the initial impact parameter h for this problem. We observe

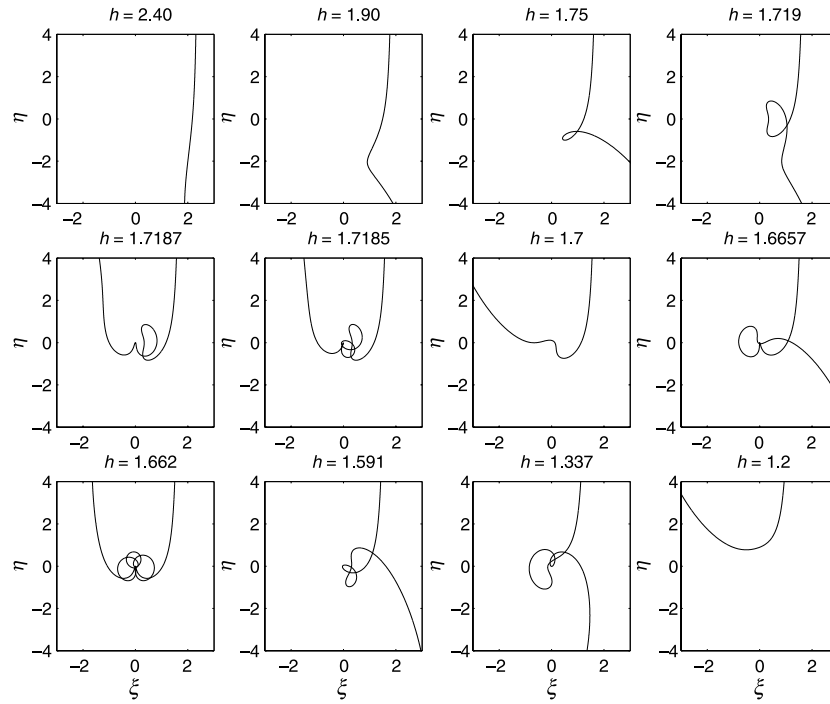


Figure 3. Orbital motion of the satellites for decreasing values of the parameter h . There are intervals of h in which small changes in this parameter implies completely different orbital motion for the interaction between the satellites.

intervals in which the dependence of h' with h is poorly resolved. In fact, for these intervals, the situation of poorly resolved dependency of h' to variations in h persists, no matter how we improve the resolution of the scale. Thus, even high magnification of these unresolved intervals does not reveal a smooth curve. These phenomenon is called *chaotic scattering* (Ott and Tél, 1993). At this point, it is important to say that for the knowledge of the author, there is no analytic proof related to the existence of a chaotic invariant set on the Hill's problem. However, recently extensive numerical works (Koon et al., 2000; Simó and Stuchi, 2000) and a phenomenological model based on the inclined billiard (Hénon, 1988; Petit and Hénon, 1989) suggest strong evidence supporting the existence of this chaotic invariant set on the range of values that we are interested in this paper. Thus, for the specific case observed in satellite encounters, and particularly for the situation related to a spacecraft exploring the moons of the Jovian system (situation that motivates this work), the model proposed by Petit and Hénon (1989) allows the introduction of a symbolic dynamics to the problem, which elucidated the dynamics and characterized it as a chaotic scattering.

Chaotic scattering is characterized by 'sensitive dependence' of output variables that characterize the particle trajectory after the scattering to small changes in an

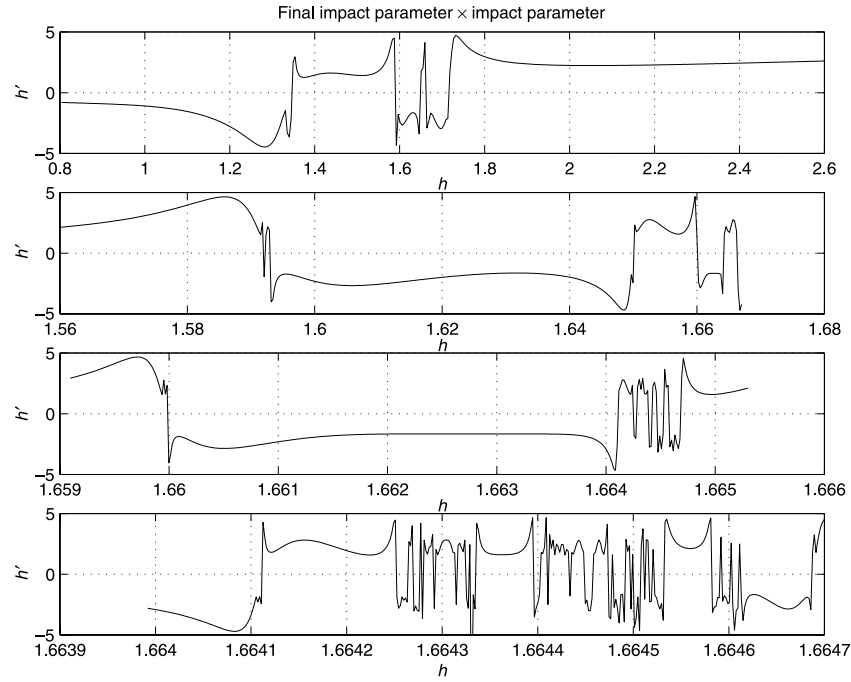


Figure 4. The scattering function which presents the relation between the final impact parameter h' and the initial impact parameter h . There are poorly resolved regions in the graphic. These regions persist even if better resolution scales are used. These behavior characterize the phenomena as a chaotic scattering.

input variable that characterizes the trajectory before scattering (Ott and Tél, 1993). The dynamics of scattering can be explained by the existence of a saddle chaotic invariant set (Hsu et al., 1988), formed by the intersection of its stable and unstable manifolds, where the stable and unstable manifolds each consist of a Cantor set of roughly parallel surfaces. Embedded in this chaotic invariant set there is an infinite, numerable and dense set of unstable periodic orbits. By using the control of chaos methods, these orbits can be used to control the orbit of one satellite around the other, as shown in the next section.

5. Controlling a Satellite Encounter

To apply our method, we introduce a Poincaré section $\eta = 0$, mapping the (ξ, u) -plane to itself, whenever the trajectory traverses the Poincaré section with $v < 0$. Thus, the Poincaré return map for fixed Γ defined is a two-dimensional map.

The SD method, in association with the NR method were used to find periodic orbits for values of the impact parameter for which the chaotic scattering is present. The first method was applied to each point of a set of starting points defined by

a uniform grid of initial conditions over the state space. For each point of this set, the method converged to a solution. For distinct starting points the method may converge to the same solution, which is particularly true if the number of the points of the grid is larger than the number of unstable periodic orbits embedded in the chaotic invariant set. Furthermore, this process of scanning the state space by using the points of the grid was repeated for each value of period of periodic orbits which we were interested in finding out. From the application of this method, we got the solution points of the desired periodic orbits and also points of quasi-periodic orbits. All these points were then used as the starting points for the NR method. This method not only allows a refinement of the points that belong to periodic orbits, but also discharges eventual points that belong to quasi-periodic orbits. Some interesting periodic orbits that were found using this procedure are shown in Figure 5.

Trajectories very close to an unstable periodic orbit follow it for a while, after which they escape from it by following the unstable manifold of the periodic orbit (Macau, 1998, 1999). Thus, in our problem situation, one satellite can be kept in orbit around the other following an unstable periodic orbit just if a control algorithm for stabilization were applied. As we are considering a situation for which the system presents a chaotic scattering, which means that the unstable periodic orbit is embedded in a nonattractive chaotic invariant set, we can apply the discussed

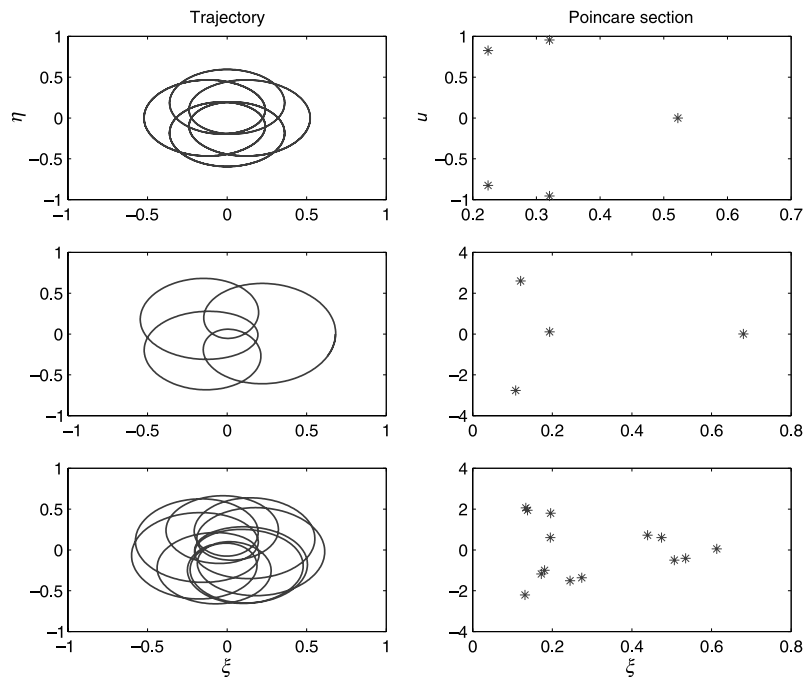


Figure 5. Some unstable periodic orbits found for $h = 1.49043$.

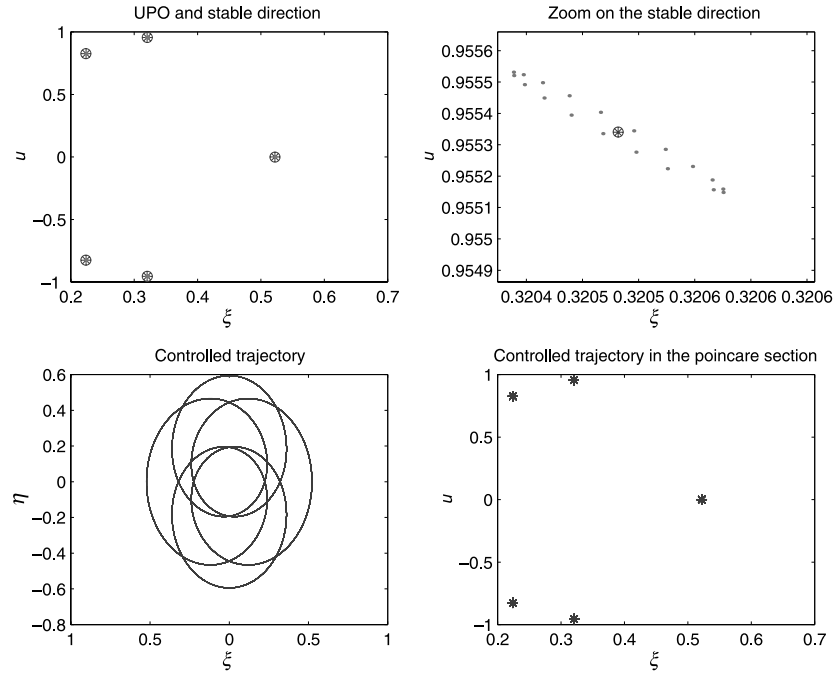


Figure 6. In the clockwise direction we have: The unstable periodic orbit embedded in the chaotic invariant set that was chosen as a desired orbit. The stable direction found by using a geometric approach (compare with Figure 1), the controlled orbit, the controlled orbit in the Poincaré section.

control of chaos technique. The result of applying this stabilization strategy appears in Figure 6. In this figure, it is possible to see how the geometric approach is used to determine the stable direction (see the second picture, starting from the top, left to right, and compare it with the one shown in Figure 1). If corrected perturbations are applied each time the trajectory crosses the defined Poincaré section as a change on the velocity (u coordinate) all of them are smaller than 0.001, considering the used normalized units. However, at this point, other approach can be used. It is possible to define a maximum allowed deviation from the desired unstable periodic orbit and just apply the corrected perturbation if the trajectory goes beyond this allowed limit. We found that both of these approaches work fine and each one to be used is decision to be made according to the rate with which the trajectory deviates from the desired unstable periodic orbit.

6. Conclusion

The purpose of this work was to show that the chaos control techniques can be used to efficiently keep a spacecraft around another body performing elaborate orbits when chaos is present. Control of chaos technique based on the OGY method can be extended to Hamiltonian situation using the concepts of stable and unstable

directions calculated in the trajectory. Using this method, we showed that the dense and infinite set of unstable periodic orbits that are embedded in the chaotic invariant set can be seen as an important source of orbits that can be exploited for astrodynamics applications. Finding the location of these unstable periodic orbits in a Hamiltonian system is not a trivial task. However, we have shown that the combined use of the SD method with the NR methods perform an efficient and powerful procedure to find unstable periodic orbits of any period.

Previous works have shown presented methods that allows us to explore the sensitive dependency on initial conditions to guide a spacecraft when chaos exists. These methods can be combined with the one presented here for chaos control so that a spacecraft can be guided to perform elaborate and complex trajectories by combining parts of unstable periodic orbits. Thus, we can choose from the periodic orbits the ones to be combined to result desired complex orbits. Then, chaos control is used to keep the spacecraft about the parts of the unstable periodic orbits of interest, while low energy transfer methods can be used to guide the spacecraft among these unstable periodic orbits. We are preparing a study that explores this strategy in astrodynamics.

Acknowledgements

This work was supported by the CNPq (Brazilian Agency for Research and Technological Development) and by FAPESP. The author would like to thank Profs Celso Grebogi, Edward Belbruno and Iberê Luiz Caldas for helpful discussions.

References

- Arecchi, F. T., Meucci, R., Puccioni, G. and Tredicce, J.: 1982, 'Experimental evidence of subharmonic bifurcations, multistability, and turbulence in a Q-switched gas laser', *Phys. Rev. Lett.* **49**, 1217.
- Arrowsmith, D. K. and Place, C. M.: 1994, *An Introduction to Dynamical Systems*, Cambridge.
- Auerbach, D.: 1990, *Phys. Rev. A* **41**, 6692.
- Azevedo, A. and Rezende, S. M.: 1991, *Phys. Rev. Lett.* **66**, 1342–1345.
- Bello-Mora, M., Graziani, F., Teofilatto, P., Circi, C., Porfilo, M. and Hechler, M.: 2000, A Systematic Analysis on Weak Stability Boundary Transfers to the Moon, *Proceedings of the 51st Inter. Astronautical Congress* **IAF-00-A.6.03**.
- Belbruno, E.: 1994, *J. Br. Interplanetary Soc.* **47**, 73.
- Belbruno, E.: 2002a, 'Analytic estimation of weak stability boundaries and low energy transfers', *Contemp. Math. Am. Math. Soc.* **292**, 17.
- Belbruno, E.: 2002b, Low Energy Transfers, Weak Capture and Chaos, *Proceedings of the 53rd Inter. Astronautical Congress* **IAF-02-A.6.03**.
- Belbruno, E.: 2003, *Capture Process and Chaotic Motions in Celestial Mechanics*, Princeton University Press, Princeton, NJ.
- Belbruno, E. and Miller, J.: 1993, *J. Guidance Contr. Dyn.* **4**, 4.
- Biham, O. and Wenzel, W.: 1989, *Phys. Rev. Lett.* **63**, 819.

- Bollt, E. N. and Meiss, J. D.: 1995, *Physica* **D81**, 280.
- Bollt, E. N. and Meiss, J. D.: 1996, *Phys. Lett. A* **204**, 373.
- Cvitanovic, P.: 1992, *Chaos* **2**, 1.
- Cvitanovic, P. and Eckhardt, B.: 1989, *Phys. Rev. Lett.* **63**, 823.
- Davidchack, R. L. and Lai, Y. C.: 1999, *Phys. Rev. E* **60**, 6172.
- Dermontt, S. F. and Murray, C. D.: 1981, *Icarus*, **48**, 1.
- Ditto, W. L., Rauseo, S. N. and Spano, M. L.: 1990, *Phys. Rev. Lett.* **65**, 3211.
- Dole, S. H.: 1962, *Planet. Space Sci.* **9**, 541.
- Grafinkel, A., Spano, M. L., Ditto, W. L. and Weiss, J. N.: 1992, *Science* **257**, 1230.
- Grassberger, P., Kantz, H. and Moenig, U.: 1989, *J. Phys. A* **22**, 5217.
- Grebogi, C. and Lai, Y. C.: 1997, *Syst. Contr. Lett.* **31**, 307.
- Goldreich, P. and Tremaine, S.: 1979, *Nature* **277**, 97.
- Goldreich, P. and Tremaine, S.: 1980, *Astrophys. J.* **241**, 425.
- Hansen, K.: 1995, *Phys. Rev. E* **52**, 2388.
- Hénon, M.: 1981, *Nature* **293**, 33.
- Hénon, M.: 1988, *Physica D* **33**, 132.
- Hénon, M. and Petit, J.-M.: 1986, *Celest. Mech.* **38**, 67.
- Hill, G. W.: 1878, *Am. J. Math.* **1**, 5.
- Hsu, G.-H., Ott, E. and Grebogi, C.: 1988, *Phys. Lett. A* **127**, 199.
- Hudson, J. L. and Mankin, J. C.: 1981, 'Chaos in Belousov-Zhabotinsky reaction', *J. Chem. Phys.* **74**, 6171.
- Koon, W. S., Lo, M. W., Marsden, J. E. and Ross, S. D.: 2000, *Chaos* **10**, 427.
- Lai, Y.-C., Ding, M. and Grebogi, C.: 1993a, *Phys. Rev. E* **47**, 86.
- Lai, Y.-C., Tél, T. and Grebogi, C.: 1993b, *Phys. Rev. E* **48**, 709.
- Libchaber, A. and Maurer, J.: 1982, A Rayleigh Bénard experiment: helium in a small box, *Proc. of the NATO Advanced Studies Institute on Nonlinear Phenomena at Phase Transitions and Instabilities*, Vol. 259.
- Macau, E. E. N.: 1998, *Phys. Rev. E* **57**, 5337.
- Macau, E. E. N.: 1999, Using Chaos to Control a Satellite Encounter, *Proceedings of the 50th Inter. Astronautical Congress IAF-99-A.1.06*.
- Macau, E. E. N.: 2000, *Acta Astronautica* **47**, 871.
- Ott, E. and Tél, T.: 1993, *Chaos* **3**, 417.
- Ott, E., Grebogi, C. and Yorke, J. A.: 1990, *Phys. Rev. Lett.* **64**, 1196.
- Petit, J.-M. and Hénon, M.: 1986, *Icarus* **66**, 536.
- Petit, J.-M. and Hénon, M.: 1989, On a Cantor structure in a satellite scattering problem, *Dynamics and Stochastic Process*, Springer Verlag, Berlin.
- Poincaré, H.: 1889, *Les Méthodes Nouvelles de la Mécanique Céleste*, Gauthier-Villars, Paris.
- Roth, B. A. F., Macau, E. E. N. and Terro, M. O.: 2003, To be published.
- Schmelcher, P. and Diakonov, F. K.: 1998, *Phys. Rev. E* **57**, 2739.
- Schofield, N.: 1981, *Mon. Not. R. Astr. Soc.* **197**, 1031.
- Shinbrot, T., Ott, E., Grebogi, C. and Yorke, J. A.: 1990, *Phys. Rev. Lett.* **65**, 3215.
- Simó, C. and Stuchi, T. J.: 2000, *Physica D* **140**, 1.